CSE STUDY HUB

Generative AI & Prompt Engineering

UNIT 2 · GPT Models and Conversational AI

What this unit is about

This unit builds understanding of **GPT Models and Conversational AI**. These notes are designed for quick learning and exam revision: learn the meaning, understand the working, practise an example, and finish with comparisons and key terms.

Syllabus coverage checklist

1. GPT meaning and autoregressive transformer approach

2. Tokenisation, embeddings, attention and context windows

3. Pre-training, instruction tuning and human-feedback alignment

4. Text generation, summarisation, translation and question answering

5. Dialogue systems: turns, context, roles and conversation state

6. Chatbot design: intent, response, fallback and escalation

7. Model parameters: temperature and output length at a conceptual level

8. Evaluation: factuality, relevance, coherence, safety and latency

How to study: Read each topic once for meaning, underline definitions and formulas, reproduce diagrams/steps without looking, then answer the revision questions on Page 5.

Exam answer pattern

For a 5-mark answer: (1) definition, (2) labelled diagram or formula, (3) working/components in steps, (4) example/application, and (5) one advantage, limitation or comparison.

UNIT 2 NOTES · PART 1

GPT meaning and autoregressive transformer approach

Start with a precise definition of GPT meaning and autoregressive transformer approach. Explain its purpose, the problem it solves, its main components and where it is used. In an exam answer, add one practical example and distinguish it from the nearest related concept.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Tokenisation, embeddings, attention and context windows

Tokenisation, embeddings, attention and context windows is a syllabus concept in Generative AI & Prompt Engineering. Learn its definition, purpose, key terms and working principle. Organise the explanation into components or steps, then add a practical example, one advantage, one limitation and a comparison with a related concept.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Pre-training, instruction tuning and human-feedback alignment

Pre-training, instruction tuning and human-feedback alignment is a syllabus concept in Generative AI & Prompt Engineering. Learn its definition, purpose, key terms and working principle. Organise the explanation into components or steps, then add a practical example, one advantage, one limitation and a comparison with a related concept.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Connect and apply

For a practical question on GPT Models and Conversational AI, first identify which syllabus concept is being tested. State assumptions, draw the relevant structure/flow, work through the steps in order, and finish by interpreting the result. Mention complexity, reliability, accuracy, cost or ethical impact whenever it is relevant.

UNIT 2 NOTES · PART 2

Text generation, summarisation, translation and question answering

Text generation, summarisation, translation and question answering is a syllabus concept in Generative AI & Prompt Engineering. Learn its definition, purpose, key terms and working principle. Organise the explanation into components or steps, then add a practical example, one advantage, one limitation and a comparison with a related concept.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Dialogue systems: turns, context, roles and conversation state

Dialogue systems: turns, context, roles and conversation state is a syllabus concept in Generative AI & Prompt Engineering. Learn its definition, purpose, key terms and working principle. Organise the explanation into components or steps, then add a practical example, one advantage, one limitation and a comparison with a related concept.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Chatbot design: intent, response, fallback and escalation

Chatbot design: intent, response, fallback and escalation is a syllabus concept in Generative AI & Prompt Engineering. Learn its definition, purpose, key terms and working principle. Organise the explanation into components or steps, then add a practical example, one advantage, one limitation and a comparison with a related concept.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Connect and apply

For a practical question on GPT Models and Conversational AI, first identify which syllabus concept is being tested. State assumptions, draw the relevant structure/flow, work through the steps in order, and finish by interpreting the result. Mention complexity, reliability, accuracy, cost or ethical impact whenever it is relevant.

UNIT 2 NOTES · PART 3

Model parameters: temperature and output length at a conceptual level

Treat Model parameters: temperature and output length at a conceptual level as a structured model. Name its components in order, explain the responsibility of each component, and show how information moves between them. Include a labelled block diagram and one use case.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Evaluation: factuality, relevance, coherence, safety and latency

Evaluation: factuality, relevance, coherence, safety and latency is a syllabus concept in Generative AI & Prompt Engineering. Learn its definition, purpose, key terms and working principle. Organise the explanation into components or steps, then add a practical example, one advantage, one limitation and a comparison with a related concept.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Connect and apply

For a practical question on GPT Models and Conversational AI, first identify which syllabus concept is being tested. State assumptions, draw the relevant structure/flow, work through the steps in order, and finish by interpreting the result. Mention complexity, reliability, accuracy, cost or ethical impact whenever it is relevant.

QUICK REVISION & EXAM PRACTICE

Key takeaways

• **GPT meaning and autoregressive transformer approach** — know the definition, core working and one example.

• **Tokenisation, embeddings, attention and context windows** — know the definition, core working and one example.

• **Pre-training, instruction tuning and human-feedback alignment** — know the definition, core working and one example.

• **Text generation, summarisation, translation and question answering** — know the definition, core working and one example.

• **Dialogue systems: turns, context, roles and conversation state** — know the definition, core working and one example.

• **Chatbot design: intent, response, fallback and escalation** — know the definition, core working and one example.

• **Model parameters: temperature and output length at a conceptual level** — know the definition, core working and one example.

• **Evaluation: factuality, relevance, coherence, safety and latency** — know the definition, core working and one example.

Worked-answer framework

Scenario: A question asks you to explain or apply a concept from GPT Models and Conversational AI. **Step 1:** identify the relevant term and define it. **Step 2:** write the formula, components or ordered process. **Step 3:** apply it to the given data/scenario. **Step 4:** state the result and justify it. **Step 5:** mention one limitation or precaution.

Self-test questions

Q1. Define GPT meaning and autoregressive transformer approach and explain its importance.

Q2. Explain the working/steps involved in Tokenisation, embeddings, attention and context windows.

Q3. Differentiate between two related concepts from GPT Models and Conversational AI.

Q4. Give one practical application of Evaluation: factuality, relevance, coherence, safety and latency and discuss one limitation.

Q5. Write a structured 5-mark answer covering the main ideas of GPT Models and Conversational AI.

Final checklist: I can define every topic · explain the process without notes · reproduce formulas/diagrams · solve one example · compare related concepts.